

PH4209 Assignment — 05

22MS037

Problem 1 — Cooperators and Defectors

The prisoner's dilemma is a game where two players can either cooperate or defect. The payoff matrix is given by:

$$M = \begin{pmatrix} 4 & 1 \\ 6 & 2 \end{pmatrix} \quad (1)$$

where the first row corresponds to the payoffs for a cooperator and the second row corresponds to the payoffs for a defector. We consider a population of $N = 1000$ individuals, where 500 are cooperators and 500 are defectors.

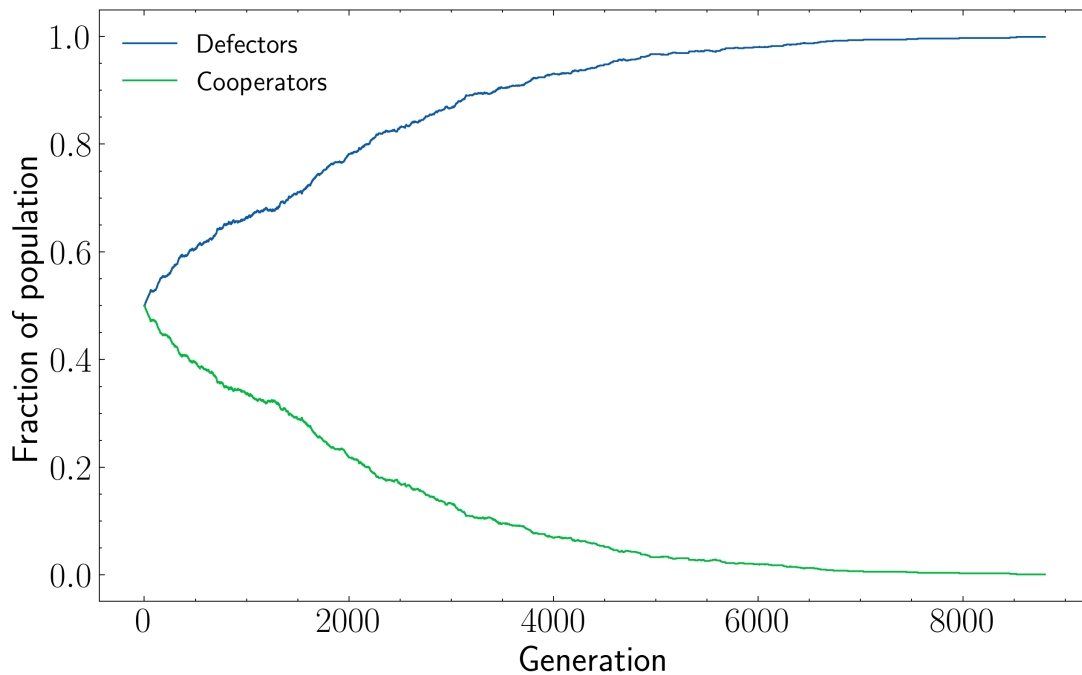


Figure 1: Fraction of cooperators and defectors over generations.

Problem 2 — TFT vs AIID

Now, we consider a population of $N = 1000$ individuals, where each individuals are either playing TFT or AIID. The payoff matrix is the same as before. We want to see how the frequencies of TFT and AIID evolve over generations.

We have two initial conditions:

1. Initially, 250 individuals are playing TFT and 750 are playing AIID.

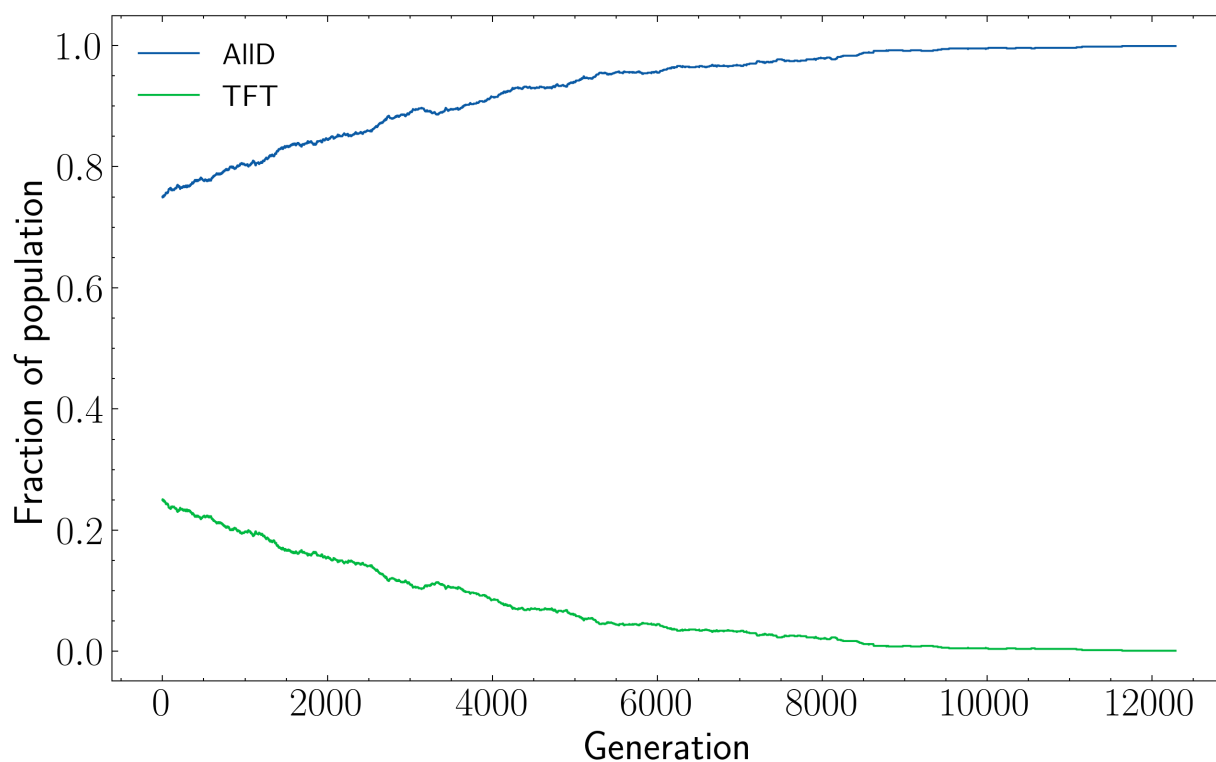


Figure 2: Fraction of TFT and AIID over generations.

2. Initially, 150 individuals are playing TFT and 850 are playing AIID.

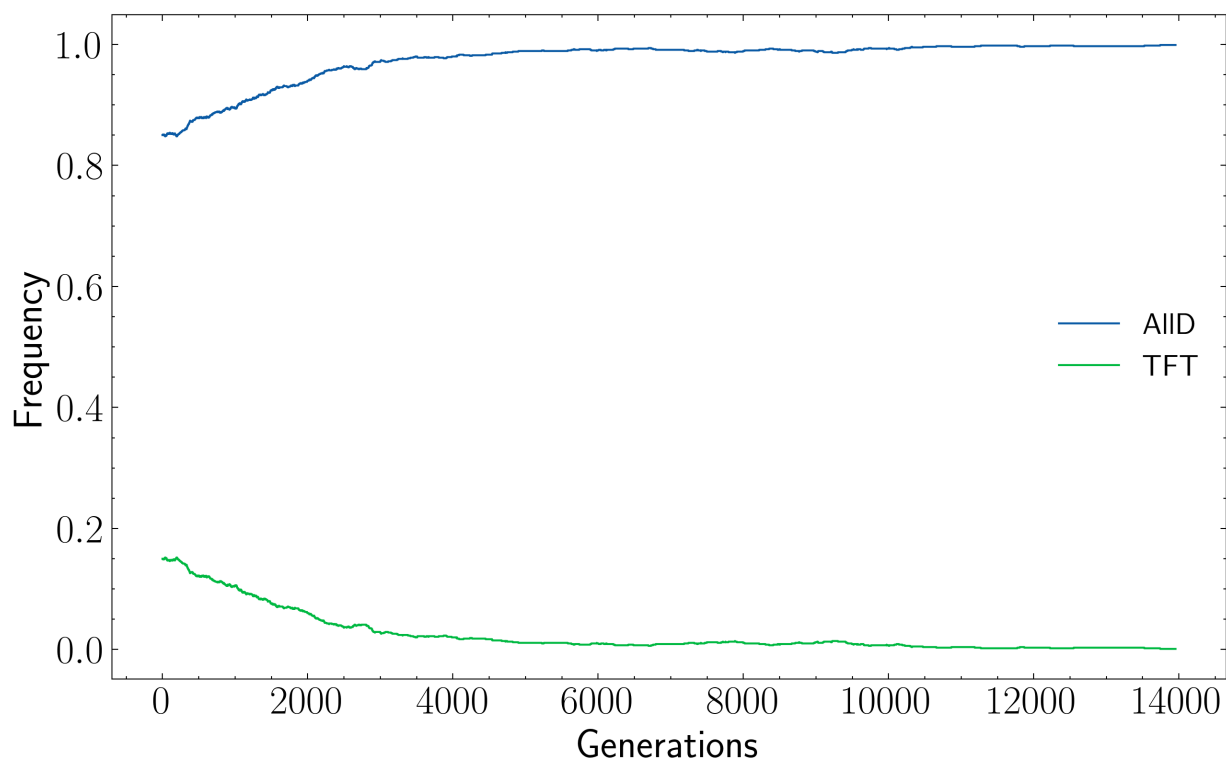


Figure 3: Fraction of TFT and AIID over generations.

One can find the average expected payoff for TFT and ALLD as,

	TFT	ALLD
TFT	ma	$b + (m - 1)d$
ALLD	$c + (m - 1)d$	md

If only two players are playing this game for m rounds, then there is an equilibrium frequency of TFT given by,

$$x = \frac{d - b}{m(a - d) - (b + c) + 2d} \quad (2)$$

For our values, it comes out to be $x = 0.2$. However, in both cases TFT goes extinct. This can be explained by how the simulation is performed.

Suppose we select two players from the population, one playing $\text{TFT}(P_1)$ and the other playing $\text{ALLD}(P_2)$. They play the game for m rounds and get payoffs f_1 and f_2 respectively. Now, suppose during some other generation we select P_1 and P_2 again. For the same number of rounds, P_1 will always get the same payoff f_1 and P_2 will always get the same payoff f_2 . This is not the natural extension from 2 players to N players. We would expect that if P_1 and P_2 play the game multiple times, they should have a memory of the previous interactions and their payoffs should change accordingly. Thus every generation, P_1 and P_2 get payoffs f_1 and f_2 respectively, instead of the payoff of P_1 increasing and the payoff of P_2 decreasing.

As a result, if the number of generations is more than some threshold, players playing each other again increases. This is why TFT goes extinct in the simulation. If we actually give them memory of the previous interactions, then TFT should be able to survive and even dominate the population.

Since, this is not clarified in the question, we have performed the simulation as described in the question. However, if we give them memory of the previous interactions, then TFT should be able to survive and even dominate the population.